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CSE4078 Introduction to Natural Language Processing Project

Delivery#1 Report

Project Task

Conversation

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Project Task Explanation

In this task related to Natural Language Processing (NLP), our objective is to develop algorithms and models that enable computers to understand, interpret, and generate human language in conversations. This involves a range of challenges, including text classification, sentiment analysis, named entity recognition, language translation, and dialogue generation. By harnessing NLP techniques, we aim to enhance communication between humans and machines, enabling more efficient and intuitive interactions across various applications such as virtual assistants, chatbots, sentiment analysis tools, and language translation services.

Datasets Explanation

We found 4 numbers of datasets for the project task. You can find more detailed information about the datasets below.

Overview Table

In this section, you can review the Table 1.1 below to get a general idea about the datasets. Later, detailed information will be given for each dataset.

#	Dataset Name	Source	URL	Description	Number of Instances
1	Turkish Reading Comprehension Question Answering Dataset	Open Source	Link	Questions and answers related to Ottoman history	6234
2	Chatbot Arena Conversations	Open Source	Link	This dataset contains chatbot questions and answers	33000
3	Turkish NLP Q&A Dataset	Open Source	Link	This dataset contains question-answer pairs in Turkish	681 instances each having 5-10 question and answers

				language.	
4	Mental Health Counseling Conversations	Hugging Face	Link	A collection of question and answers sourced from two online counseling and therapy platforms.	3510

Table 1.1. Overview of Datasets

Dataset#1

General Information

Name: Turkish Reading Comprehension Question Answering Dataset

Source: Open Source

Number of Instances: 6234

Description

This dataset is created for Türkiye Açık Kaynak Platformu. It contains 6234 questions and answers related to Ottoman history.

Example Instance

```
{
  "question": "Bursayı 1326 yılında kim kuşatmıştır?",
  "id": 15,
  "answers": [
    {
      "answer_start": 369,
      "text": "Osman Bey,"
    }
  ]
}
```

Example Instance in Alpaca Format

```
{
  "instruction": "Answer the question",
  "input": "Bursayı 1326 yılında kim kuşatmıştır?",
```

```
    "output": "Osman Bey,",  
}
```

Dataset#2

General Information

Name: Chatbot Arena Conversations

Source: Hugging Face

Number of Instances: 33000 instances each having 2 questions and answers.

Description

This dataset contains 33K cleaned conversations with pairwise human preferences. It is collected from 13K unique IP addresses on the Chatbot Arena from April to June 2023. Each sample includes a question ID, two model names, their full conversation text in OpenAI API JSON format, the user vote, the anonymized user ID, the detected language tag, the OpenAI moderation API tag, the additional toxic tag, and the timestamp.

Example Instance

```
{  
  "question_id" : 2564acd09e3942fd97657d05282d4389  
  "model_a" : oasst-pythia-12b  
  "model_b" : alpaca-13b  
  "winner": tie  
  "judge": arena_user_973  
  "conversation_a" : [  
    { "content": "Why did my parent not invite me to their  
wedding?", "role": "user" },  
    { "content": "It is possible that your parent did not  
invite you to their wedding for a variety of reasons. It  
could be that they did not want to burden you with the  
planning or expense of the event, or they may not have  
had the time or resources to include you. Additionally,  
it could be that they did not feel comfortable having  
you attend due to your age or relationship with them. It  
is important to remember that your parent may have had  
their own reasons for not inviting you, and it is not  
your place to judge their choices. If you feel neglected  
or upset about not being invited, it is best to talk to  
them about it and express your feelings in a  
non-accusatory way.", "role": "assistant" } ]  
  "conversation_b" : [  
    { "content": "Why did my parent not invite me to their  
wedding?", "role": "user" },
```

```

    { "content": "It is likely that they wanted to keep the
    guest list small and intimate. Maybe they felt that
    inviting you would make the event too crowded, or that
    they wanted to keep the event reserved for close friends
    and family. It could be that they wanted to avoid any
    potential drama or conflict that could arise from having
    too many people at the event. Whatever the reason, it's
    important to remember that they made this decision out
    of love and respect for you, and that their decision
    doesn't define your relationship.", "role": "assistant"
    }
  ]
  "turn": 1
  "anony": true
  "language": English
  "tstamp" 1,682,351,591.1322
  "openai_moderation" : { ...}
  "toxic_chat_tag" : { "... "}
}

```

Example Instance in Alpaca Format

```

{
  "instruction": "Answer the question",
  "input": "Why did my parent not invite me to their wedding?",
  "output": "It is possible that your parent did not invite you
  to their wedding for a variety of reasons. It could be
  that they did not want to burden you with the planning or
  expense of the event, or they may not have had the time or
  resources to include you. Additionally, it could be that
  they did not feel comfortable having you attend due to
  your age or relationship with them. It is important to
  remember that your parent may have had their own reasons
  for not inviting you, and it is not your place to judge
  their choices. If you feel neglected or upset about not
  being invited, it is best to talk to them about it and
  express your feelings in a non-accusatory way.,"
}

```

Dataset#3

General Information

Name: Turkish NLP Q&A Dataset

Source: Open Source

Number of Instances: 681

Description

This dataset is a Turkish Q&A dataset on the subject of Turkish-Islamic Science History, prepared for the Teknofest 2018 Artificial Intelligence competition.

Example Instance

```
{
  "title": "Mustafa Behçet Efendi",
  "paragraphs": [
    {
      "qas": [
        {
          "question": "Mustafa Behçet Efendi'nin mesleği nedir?",
          "id": 1001,
          "answers": [
            {
              "answer_start": 24,
              "text": "Osmanlı saray hekimi"
            }
          ]
        },
        {
          "question": "Mustafa Behçet Efendi'nin açılmasında emeği geçtiği şeyin adı nedir?",
          "id": 1002,
          "answers": [
            {
              "answer_start": 141,
              "text": "Tıbbiye"
            }
          ]
        }
      ]
    },
    {
      "context": "Mustafa Behçet Efendi , Osmanlı saray hekimidir. III. Selim ve II. Mahmut devirlerinde Osmanlı saray hekimliği yapan Mustafa Behçet Efendi, Tıbbiye'nin kurulmasında emek vermiş; tıp ve tabiyat bilimlerinin önemli eserlerinden bazılarını Türkçeye çevirmiştir. Onun bu çalışmaları o devirde Osmanlıla'da hâkim olan tıp ve biyoloji anlayışı dışına çıkma eğilimi gösteren ilk çalışmalar idi."
    }
  ]
}
```

Example Instance in Alpaca Format

```
{
  "instruction": "Generate a continuation for the following
conversation.",
  "input": {
    "conversation_id": "12345",
    "context": [
      {
        "speaker": "User",
        "utterance": "Mustafa Behçet Efendi'nin mesleği nedir?"
      },
      {
        "speaker": "Bot",
        "utterance": "Mustafa Behçet Efendi Osmanlı saray hekimi
idi."
      },
      {
        "speaker": "User",
        "utterance": "Onun önemli bir katkısı oldu mu?"
      }
    ]
  },
  "output": {
    "speaker": "Bot",
    "continuation": "Evet, Mustafa Behçet Efendi'nin Tıbbiye'nin
kurulmasında önemli bir katkısı oldu."
  }
}
```

Dataset#4

General Information

Name: Mental Health Counseling Conversations

Source: HuggingFace

Number of Instances: 3510

Description

This dataset is a collection of question and answers sourced from two online counseling and therapy platforms. The language of the dataset is English but we are planning to translate the instances to Turkish. Format of the dataset is CSV.

Example Instance

```
{
  "questionID": "0",
  "questionTitle": "Can I change my feeling of being worthless
    to everyone?",
  "questionText": "I'm going through some things with my
    feelings and myself. I barely sleep and I do nothing
    but think about how I'm worthless and how I shouldn't
    be here.
    I've never tried or contemplated suicide. I've always
    wanted to fix my issues, but I never get around to it.
    How can I change my feeling of being worthless to
    everyone?",
  "questionLink":
    "https://counselchat.com/questions/can-i-change-my-feel-
    ing-of-being-worthless-to-everyone",
  "topic": "depression",
  "therapistInfo": "Sherry Katz, LCSWCouples and Family
    Therapist, LCSW",
  "therapistURL":
    "https://counselchat.com/therapists/sherry-katz-lcsw",
  "answerText": "If everyone thinks you're worthless, then
    maybe you need to find new people to hang out
    with.Seriously, the social context in which a person
    lives is a big influence in self-esteem.Otherwise, you
    can go round and round trying to understand why you're
    not worthless, then go back to the same crowd and be
    knocked down again.There are many inspirational
    messages you can find in social media. Â Maybe read
    some of the ones which state that no person is
    worthless, and that everyone has a good purpose to
    their life.Also, since our culture is so saturated with
    the belief that if someone doesn't feel good about
    themselves that this is somehow terrible.Bad feelings
    are part of living. Â They are the motivation to remove
    ourselves from situations and relationships which do us
    more harm than good.Bad feelings do feel terrible. Â
    Your feeling of worthlessness may be good in the sense
    of motivating you to find out that you are much better
    than your feelings today.",
  "upvotes": 1,
  "views": 2899,
  "split": "train"
}
```

Example Instance in Alpaca Format

```
{
  "instruction": "Generate a continuation for the following
conversation context.",
  "input": {
    "conversation_id": "54321",
```

```

"context": [
  {
    "speaker": "User",
    "utterance": "I've been struggling with
feelings of worthlessness and not knowing how to
change them."
  },
  {
    "speaker": "Bot",
    "utterance": "It's tough dealing with those
emotions. Have you considered seeking support or
talking to someone about how you feel?"
  },
  {
    "speaker": "User",
    "utterance": "I haven't thought about that.
Do you think changing my social circle could
help?"
  }
],
"output":
{
  "speaker": "Bot",
  "continuation": "Changing your social circle could
indeed have a positive impact on your self-esteem.
Surrounding yourself with supportive and uplifting
people can make a difference in how you perceive
yourself. Additionally, exploring inspirational messages
or seeking professional guidance can also contribute to
changing your feelings of worthlessness."
}
}

```